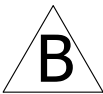


Worldwide Experts in Water TreatmentSM



LEMAY NO. 3 PUMP STATION, MO
BRACKET GREEN BOSKER
MAIN CONTROL PANEL

D SIZE SHEET (DO NOT SCALE PRINTS)					THIRD ANGLE PROJECTION		LEMAY NO. 3 PUMP STATION, MO BRACKET GREEN BOSKER MAIN CONTROL PANEL		
DRAWN BY SMM					CHECKED BY JG		DATE 10/18/2024		
PROJECT CSW1807-01					REF		THIS DRAWING CONTAINS CONFIDENTIAL PROPRIETARY INFORMATION OF OVIVO AND ITS AFFILIATES AND IS NOT TO BE DISCLOSED NOR TO BE USED EXCEPT FOR EVALUATING PROPOSALS OF OVIVO OR INSTALLING, OPERATING OR MAINTAINING OVIVO EQUIPMENT. UNLESS OTHERWISE AUTHORIZED IN WRITING BY OVIVO. UNCONTROLLED COPY IF PRINTED.		D REVISION
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1 of 21									

UPDATED AS PER COMMENTS	027014	AB	MH	04/14/26	△
UPDATE AS PER COMMENTS	025220	SMM	JW	05/14/25	△
UPDATE AS PER COMMENTS	025058	SMM	HA	04/16/25	△
INITIAL RELEASE	024172	SMM	JG	10/18/24	△
REVISION	EN/ECO	BY	CHECK'D	DATE	△

CONSTRUCTION NOTES

CONTROL PANEL SPECIFICATIONS

1.

COMPONENTS SPECIFIC

ENCLOSURE TO BE NEMA 12, 304 SS CONSTRUCTION.

ATTACH NAMEPLATES WITH MANUFACTURER'S RECOMMENDED ADHESIVE TO MAINTAIN NEMA STANDARD OF ENCLOSURE

CONTROL PANEL SHALL BE LABELED WITH SERIALIZED UL LABEL

PANEL WILL BE LABELED WITH IDENTIFICATION PLATE AS NOTED ON ENCLOSURE DRAWING.

ALL RELAY CONTACTS ARE RATED AT 10 AMPS UNLESS OTHERWISE NOTED.

COMPONENT IMAGES MAY NOT ACCURATELY REPRESENT THE ACTUAL DEVICE IN SOME INSTANCES. DIMENSIONS OF DEVICES, HOWEVER, WILL BE ACCURATE FOR PURPOSES OF LAYOUT AND SPACING.

SUBSTITUTIONS FOR COMPONENTS MAY BE PERMITTED UPON APPROVAL BY OVIVO ENGINEERS.
2.

WIRE SIZING & TYPE

WIRE SIZING SHALL BE NO LESS THAN AS FOLLOWS (UNLESS OTHERWISE NOTED):

480 3 PHASE - 14 AWG TYPE: MTW

120 1 PHASE - 16 AWG TYPE: MTW

24 VDC - 16 AWG TYPE: MTW
3.

WIRE COLOR

WIRE COLOR SHALL BE AS FOLLOWS:

POWER (480 VOLTS)

GREEN - GROUND

BLACK - ALL PHASES

CONTROL (120 VAC)

RED - LOAD WIRES FOR ALL LOCAL CONTROL CIRCUITS

WHITE - NEUTRAL FOR ALL LOCAL CONTROL CIRCUITS

YELLOW - FOREIGN POWER IN PANEL

WHITE W/ YELLOW STRIPE - NEUTRAL FOR FOREIGN POWER IN PANEL

CONTROL (24 VDC)

BLUE - LOAD WIRES FOR ALL LOCAL CONTROL CIRCUITS

WHITE W/BUE STRIPE - COMMON FOR ALL LOCAL CONTROL CIRCUITS

4.

CONNECTIONS

ALL CONNECTIONS SHALL BE POINT TO POINT WITHOUT SPLICES. EXCEPTIONS APPLY TO SOLENOIDS AND MOTORS AND FACTORY WIRED COMPONENTS.

ALL CONDUCTORS MUST BE MARKED WITH MACHINE PRINTED TAGS AT EACH TERMINATION. MARKERS SHALL BE OF HEAT SHRINK TYPE.

5.

DOCUMENTATION

FABRICATOR SHALL PROVIDE ONE COPY OF ALL DOCUMENTS SUPPLIED WITH THE COMPONENTS SUPPLIED BY THE FABRICATOR. THIS INFORMATION INCLUDES BUT IS NOT LIMITED TO MANUFACTURERS SPECIFICATIONS, INSTALLATION MANUALS AND OPERATION MANUALS.

ANY OVIVO APPROVED CHANGES OR MARK-UPS TO THE SCHEMATICS, PANEL LAYOUTS AND ANY OTHER PERTAINING DOCUMENTS SHALL BE MAINTAINED AND DELIVERED TO OVIVO AT ACCEPTANCE OF THE PANEL.

6.

TESTING

OVIVO ENGINEERS OR OVIVO APPROVED FABRICATOR, WILL PERFORM A POINT TO POINT CONTINUITY TEST ON THE PANEL.

WHEN APPLICABLE, THE FABRICATOR WILL ALSO PROVIDE POWER TO THE 480VAC OR 120VAC CIRCUITS WHERE APPLICABLE FOR VERIFICATION AND TESTING. FABRICATOR SHALL ALLOW A MINIMUM OF 2 HOURS OF ACCESS TO THE CONTROL PANEL BY ENGINEER FOR THIS AND PREVIOUSLY STATED TEST.

7.

WIRING CONNECTIONS

INTERNAL PANEL WIRING

FIELD WIRING (NOT OVIVO SCOPE)

FIELD WIRING (PROVIDED BY OVIVO, INSTALLATION BY OTHERS)

COMMUNICATION
- SIZE SHEET
DO NOT SCALE PRINTS

THIRD ANGLE
PROJECTION

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WORKMANSHIP STANDARD ERROR! APPLIES

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DWG
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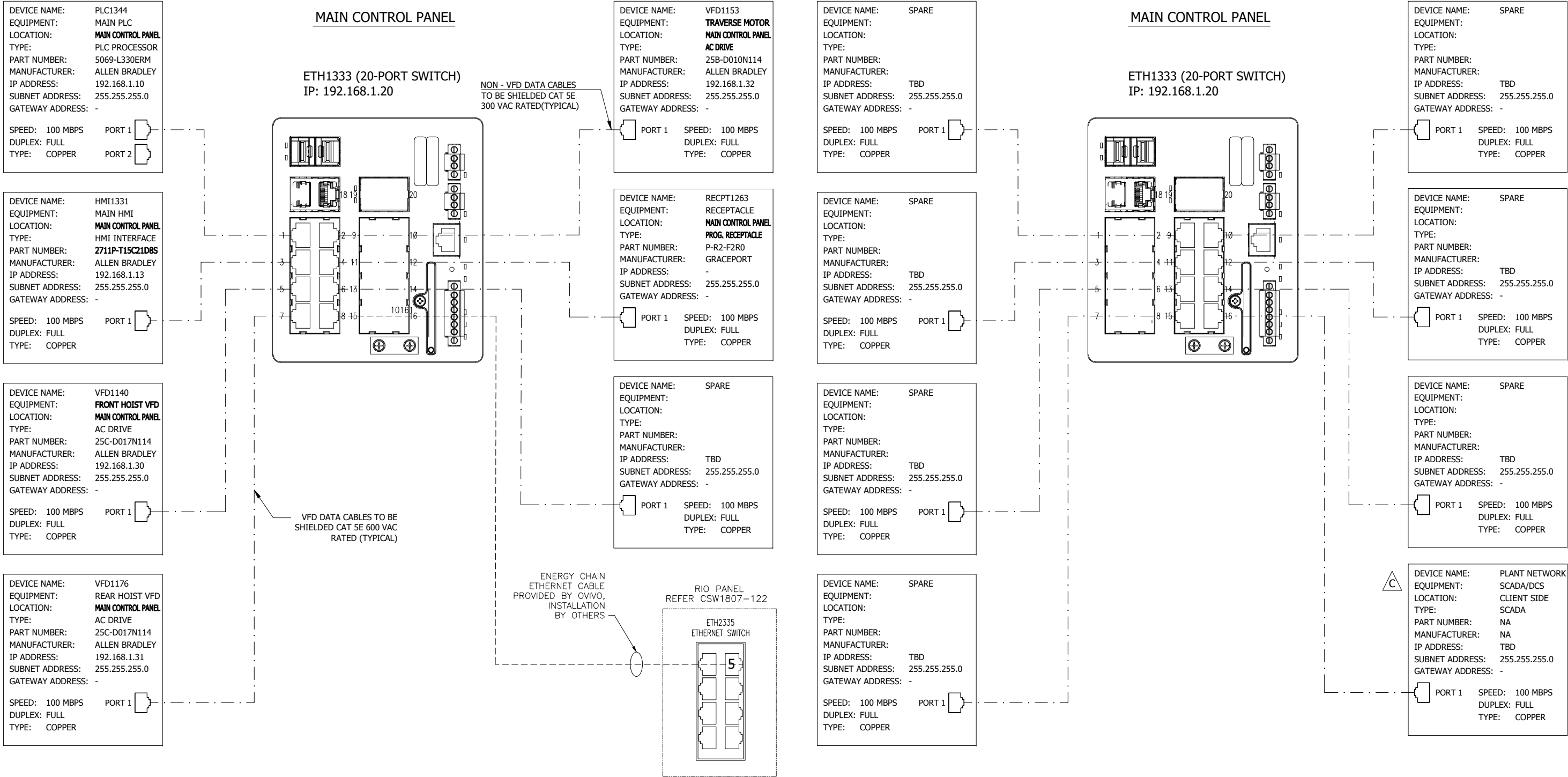
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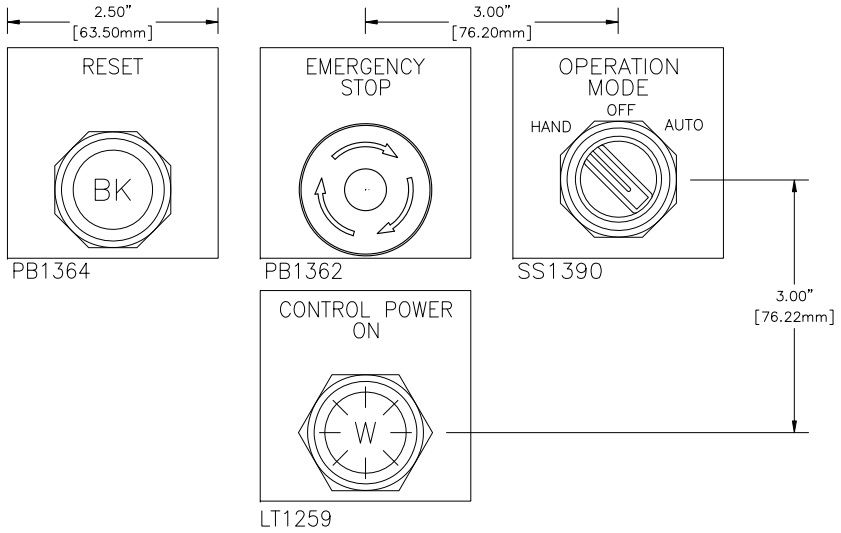
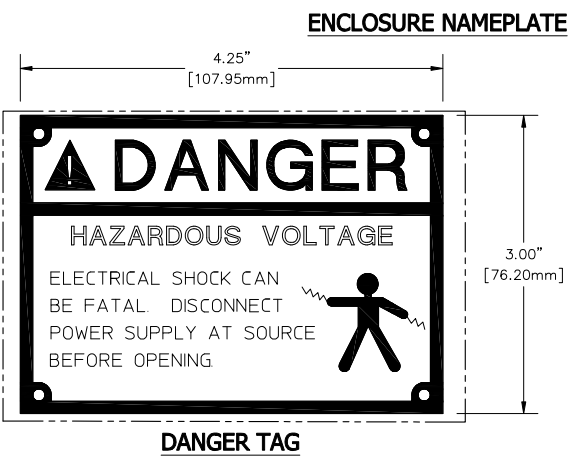
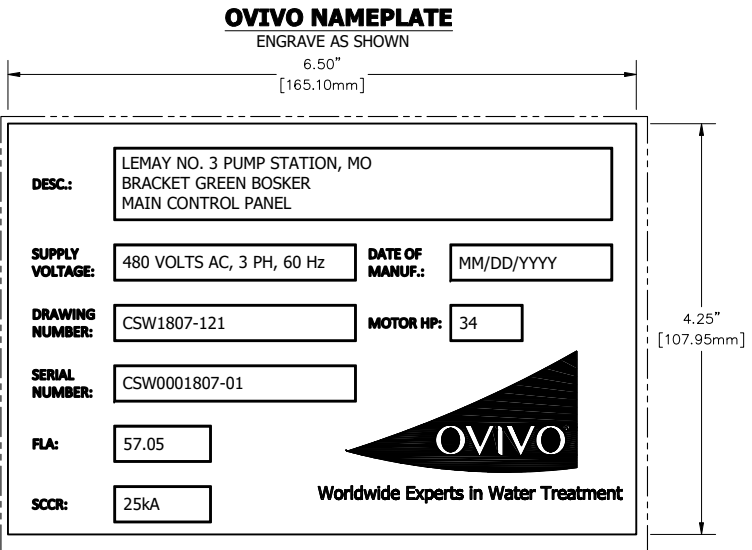
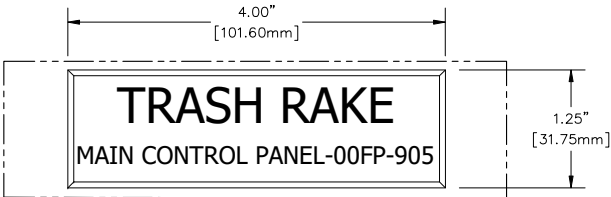
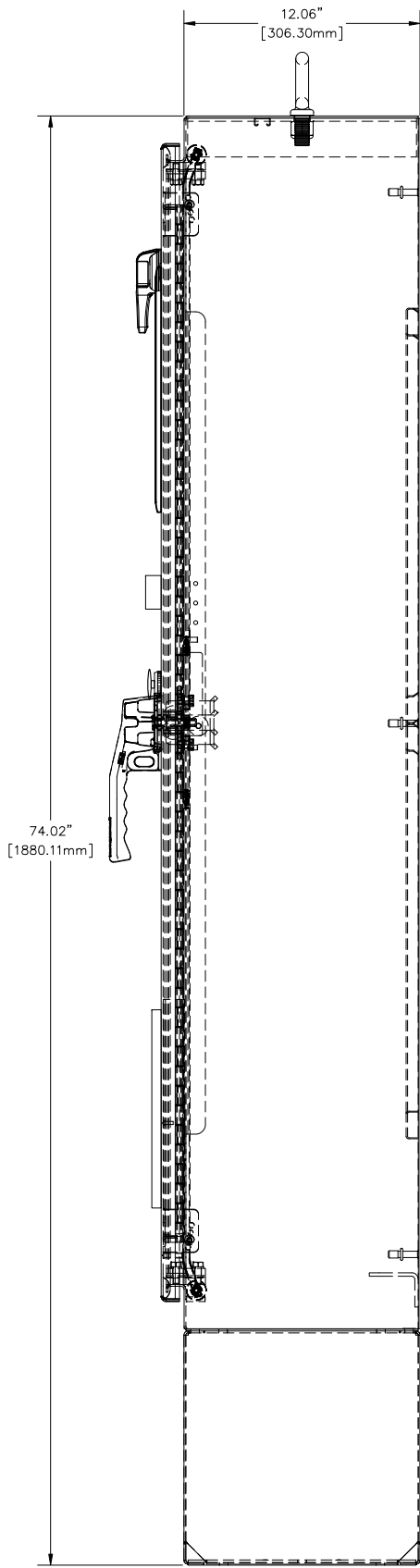
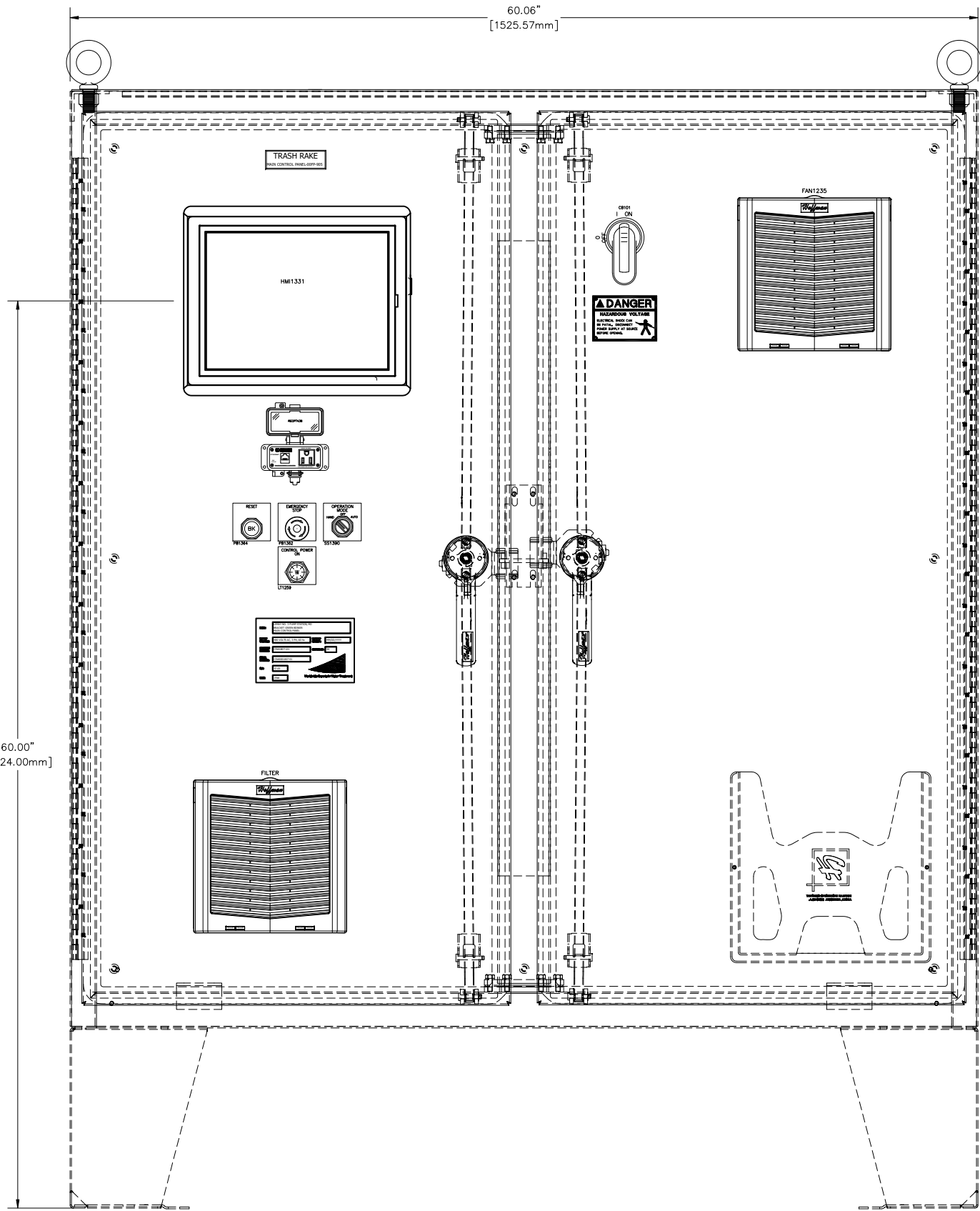
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NETWORK COMMUNICATION DIAGRAM

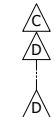


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BILL OF MATERIAL

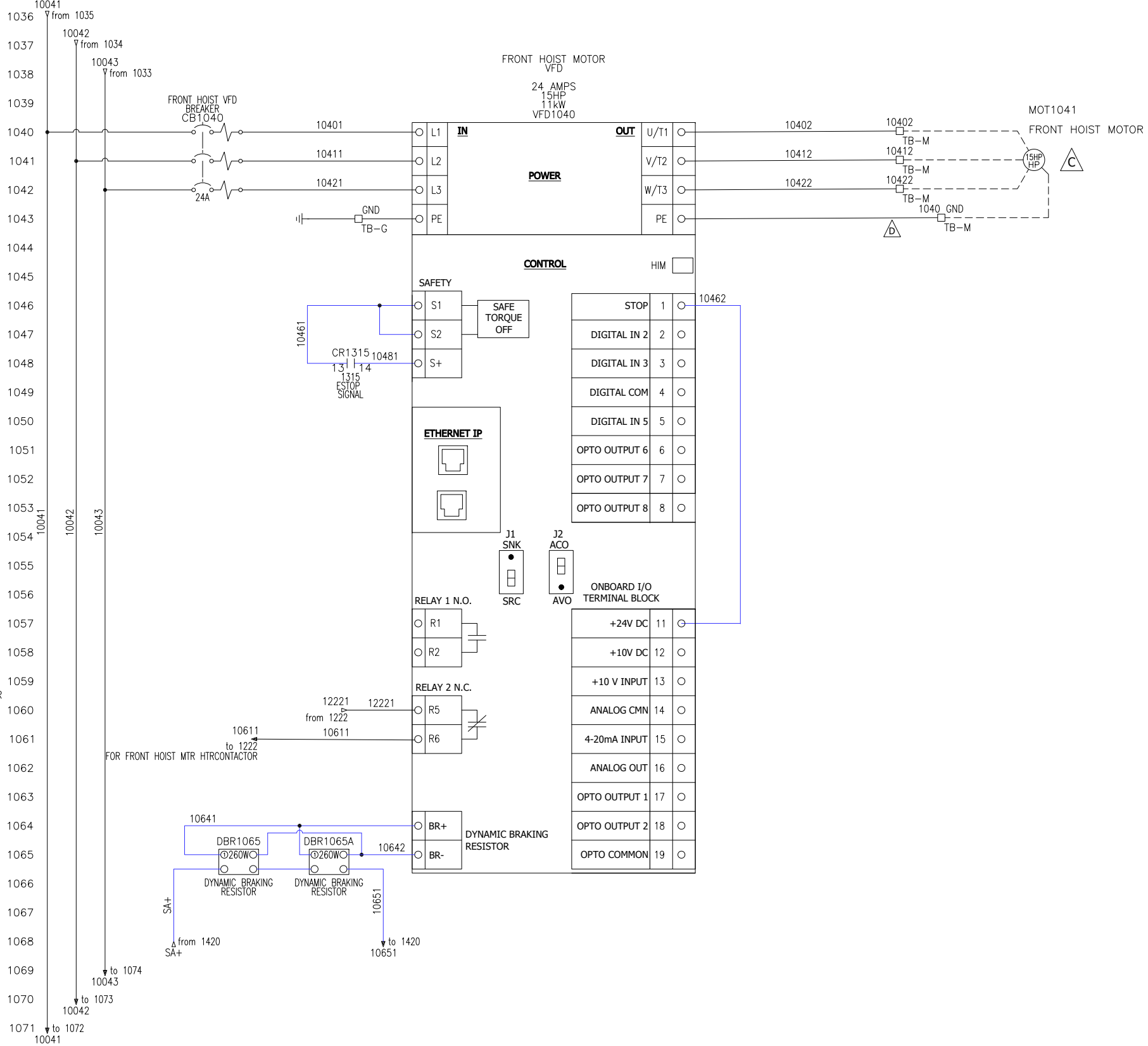
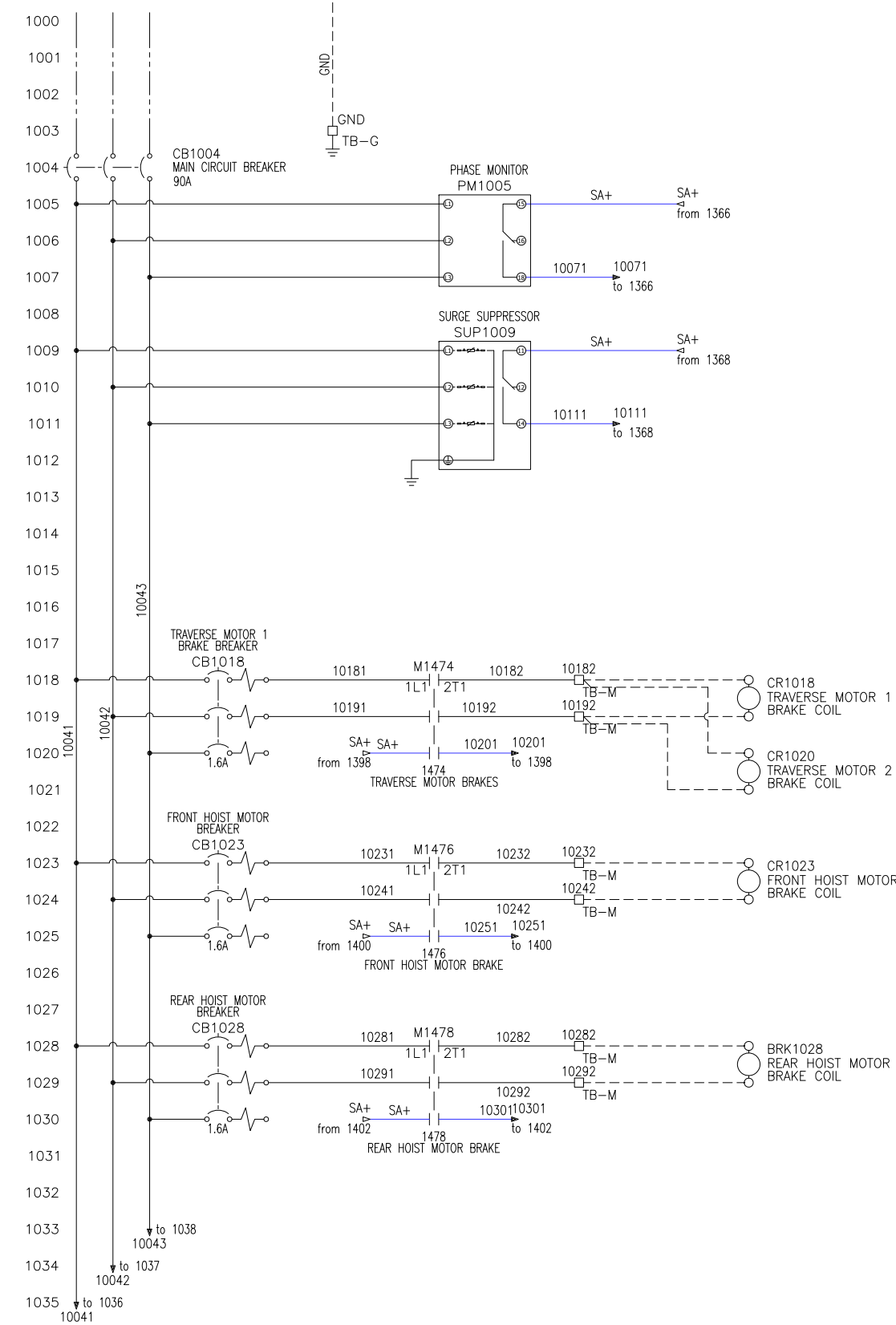
ITEM	TAGS	QTY	CATALOG	MFG	DESCRIPTION
1	ENCLOSURE	1	A62H60125SLP3PT	HOFFMAN	2-DOOR FLOOR-STAND 3-POINT LATCHES TYPE 4X, 62"X60"X12", SS TYPE 304
2	ENCLOSURE	1	A60P60	HOFFMAN	PANEL FOR TYPE 3R 4 4X 12 13 ENCLOSURE, FITS 60X48, WHITE, STEEL
3	ENCLOSURE	1	AHC12385	HOFFMAN	CORROSION INHIBITOR SPRAY, 9.45 OUNCE CAN, PACKAGE QUANTITY 6
4	FAN1235	1	HF1016414	HOFFMAN	HF10 SIDE-MOUNT FILTER FAN, 115V 159CFM, LT GRAY
5	FAN1235	1	HG1000504	HOFFMAN	CLICK IMAGE TO EXPAND INTO LARGER VIEWHOVER TO MAGNIFY IMAGE OR CLICK TO ENLARGE HG FILTER FAN EXHAUST GRILLE IP55, FITS HF10 FANS 10-INCH, LT
6	TAS1235	1	THERM16F	HOFFMAN	THERMOSTAT CONTROLLER, 2.64X1.97X1.50, 115V FAHREHNEIT, LT GRAY, PLASTIC
7	LT1255, LT1256	2	EL900D	HOFFMAN	LED ENCLOSURE LIGHT, 900 LM, 100-240 VAC
8	LT1255, LT1256	2	ELA02MF	HOFFMAN	MAGNETS FOR MAGNETIC MOUNTING
9	LT1255, LT1256	2	ELC1001PBUL	HOFFMAN	CABLE, INFEED FOR EL LED ENCLOSURE LIGHTS, 1.0 METERS, 100-240 VAC, 3-POLE, BLACK
10	LT1255, LT1256	2	ELCN1	HOFFMAN	CONNECTORS FOR EL LED ENCLOSURE LIGHT, AC, 250A, 16V, BLACK
11	RECPT1249	1	5603049	PHOENIX CONTACT	RAIL MOUNTED DUAL POWER OUTLET WITH TWO 120VAC 20A RECEPTACLES EQUIPPED WITH GROUND FAULT CIRCUIT INTERRUPTION
12	RECPT1263	1	P-R2-F2R0	GRACEPORT	PANEL INTERFACE CONNECTOR WITH CATEGORY 5E RJ45; PANEL MOUNT HOUSING; UL TYPE 4X; SIMPLEX OUTLET
13	ENCLOSURE	1	A62BP12	HOFFMAN	UNIVERSAL BARRIER PANEL FOR UNIBODY FREE-STAND, FITS A=62 AND C=12, WHITE, STEEL
14	HTR1240	1	DAH4001B	HOFFMAN	ELECTRIC HEATER, 115VAC 400W, 7.50X4.25X4.38 INCH, BRUSHED, ALUMINUM
15	CB1104	1	140G-G2C3-C90	ROCKWELL AUTOMATION	MOLDED CASE,90 AMP, 3-POLE, 600V/347 VOLT AC, G FRAME, 25 KAIC
16	CB1104	1	140G-G-RVM12B	ROCKWELL AUTOMATION	ACCESSORY FOR MCCB FOR VARIABLE DEPTH ROTARY OPERATING KIT, INCLUDES EXTERNAL HANDLE, OPERATING SHAFT, AND MCCB MOUNTED
17	CB1104	1	KA2U	BURNDY	SINGLE GROUND LUG
18	CB102	1	140G-G-TLC13	ROCKWELL AUTOMATION	TERMINAL LUGS FOR MCCB, COPPER
19	PM1005	1	2903528	PHOENIX CONTACT	PHASE MONITORING RELAY, FOR PHASE SEQUENCE AND ASYMMETRY OF 3-PHASE VOLTAGES 480VAC, 1PDT, WITH SCREW CONNECTION
20	SUP1009	1	2910386	PHOENIX CONTACT	TYPE 2 SURGE PROTECTION DEVICE, 3-CHANNEL WITH REMOTE INDICATOR CONTACT FOR 480 VAC DELTA
21	CB1018, CB1023, CB1028	3	140MT-C3E-B25	ALLEN BRADLEY	BREAKER, MOTOR PROTECTION, ROTARY STYLE, 1.6-2.5A, 480VAC, C-FRAME, MAGNETIC TRIP
22	CB1153	1	140MT-C3E-C16	ALLEN BRADLEY	BREAKER, MOTOR PROTECTION, ROTARY STYLE, 10-16A, 480VAC, C-FRAME, MAGNETIC TRIP
23	CB1040, CB1076	2	140MT-F9E-C32	ALLEN BRADLEY	BREAKER, MOTOR PROTECTION, ROTARY STYLE, 24-32A, 480VAC, F-FRAME, THERMAL/FIXED MAGNETIC TRIP
24	CB1040, CB1076, CB1118, CB1123, CB1128, CB1153	6	140MTCAFA11	ALLEN BRADLEY	140MT AUXILIARY CONTACT WITH 1NO CONTACT AND 1NC CONTACT FRONT MOUNT
25	M1474, M1476, M1478, M1486	4	100-E09KJ10	ROCKWELL AUTOMATION	100-E-MCS-E CONTACTOR, 9A, AC3 DUTY, 24-60V AC, 20-60V DC ELECTRONIC COIL, 1 N.O
26	ACCESSORY FOR MPCB	2	140MT-C-WTEN	ALLEN BRADLEY	FEEDER TERMINAL - COMPACT BUSBARS, 64 A
27	ACCESSORY FOR MPCB	2	140MT-C-W453	ALLEN BRADLEY	COMPACT BUSBAR, 64A, 3 X 45 MM SPACING, FOR 140MT, MOTOR PROTECTION
28	OL1147, OL1153	2	193-T1APM	ALLEN BRADLEY	DIN RAIL/PANEL MOUNT ADAPTER
29	OL1147, OL1153	2	193-T1AB40	ALLEN BRADLEY	RELAY, OVERLOAD, 2.9 - 4.0A, T1, IEC, BI-METALLIC
30	VFD1153	1	25B-D010N114	ALLEN BRADLEY	AC Drive, with Embedded EtherNet/IP and Safety, 380...480V AC, 3 Phase, 10.5 Amps, 5 Hp, 4 kW Normal Duty; 5 Hp, 4 kW Heavy Duty, POWERFLEX 525, SHP, Frame B, IP20 NEMA / Open Type, Filter
31	VFD1040, VFD1076	2	25C-D024N114	ALLEN BRADLEY	AC Drive, with Embedded EtherNet/IP and Safety, 380...480V AC, 3 Phase, 24 Amps, 15 Hp, 11 kW Normal Duty; 15 Hp,11 kW Heavy Duty, POWERFLEX 527, 15HP, Frame C, IP20 NEMA / Open Type, Filter
32	DBR1065, DBR1065A, DBR1101, DBR1101A	4	AK-R2-120P1K2	ALLEN BRADLEY	BRAKING RESISTOR - ENCLOSED - PowerFlex 70 DB Resistor, 120 ohm, 260 watt, PROTECTED, 260W, PROTECTED BY TEMPERATURE CONTROL SWITCH
33	CB15444, CB15448	2	1489-M1C005	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 0.5 AMPS, C-CURVE, 120VAC
34	CB1222, CB1226, CB1230, CB1243, CB1245, CB1255	6	1489-M1C010	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 1 AMPS, C-CURVE, 120VAC
35	CB1235	1	1489-M1C020	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 2 AMPS, C-CURVE, 120VAC
36	CB1263, CB1268, CB1271, CB1290	4	1489-M1C030	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 3 AMPS, C-CURVE, 120VAC
37	CB1240	1	1489-M1C050	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 5 AMPS, C-CURVE, 120VAC
38	CB1294	1	1489-M1C080	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 8 AMPS, C-CURVE, 120VAC
39	CB1249	1	1489-M1C100	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 10 AMPS, C-CURVE, 120VAC
40	CB1300, CB1306	2	1489-M1C150	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 15 AMPS, C-CURVE, 120VAC
41	CB1189A	1	1489-M1C200	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 20 AMPS, C-CURVE, 120VAC
42	CB1219	1	1489-M1C320	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 1-POLE, 32 AMPS, C-CURVE, 120VAC
43	CB1183	1	1489-M2D160	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 2-POLE, 16 AMPS, D-CURVE, 120VAC
44	CB1189	1	1489-M3D030	ALLEN BRADLEY	MINIATURE CIRCUIT BREAKER, 3-POLE, 3 AMPS, D-CURVE, 120VAC
45	XF1184	1	SP3000ACP	HAMMOND	INDUSTRIAL CONTROL TRANSFORMER, 3.0 KVA, PRIMARY 240/480 VOLTS, SECONDARY 120/240 VAC
46	XF1184	1	SPFG1	HAMMOND	FINGER GUARDS FOR TRANSFORMER
47	UPS1301/ BAT1303	1	UBC10.241	PULS	UNINTERRUPTABLE POWER SUPPLY, DC-UPS FOR BUILT-IN BATTERY 24VDC INPUT AND 24VDC OUTPUT, 10A
48	PWS1189	1	QT20.241	PULS	POWER SUPPLY, 3-PHASE AC 480VAC TO 24 VDC, 20 AMP, 480 WATT
49	ECB1330	1	2905744	PHOENIX CONTACT	8-CHANNEL, ELECTRONIC CIRCUIT BREAKER WITH ACTIVE CURRENT LIMITATION FOR PROTECTING EIGHT LOADS AT 24 V DC IN THE EVENT OF OVERLOAD AND SHORT CIRCUIT. WITH NOMINAL CURRENT ASSISTANT AND ELECTRONIC LOCKING OF THE SET NOMINAL CURRENTS. FOR INSTALLATION ON DIN RAILS.
50	PLC1344	1	5069-L330ERM	ALLEN BRADLEY	COMPACTLOGIX ETHERNET MOTION CONTROLLER, ETHERNET/IP, USB COMMUNICATION

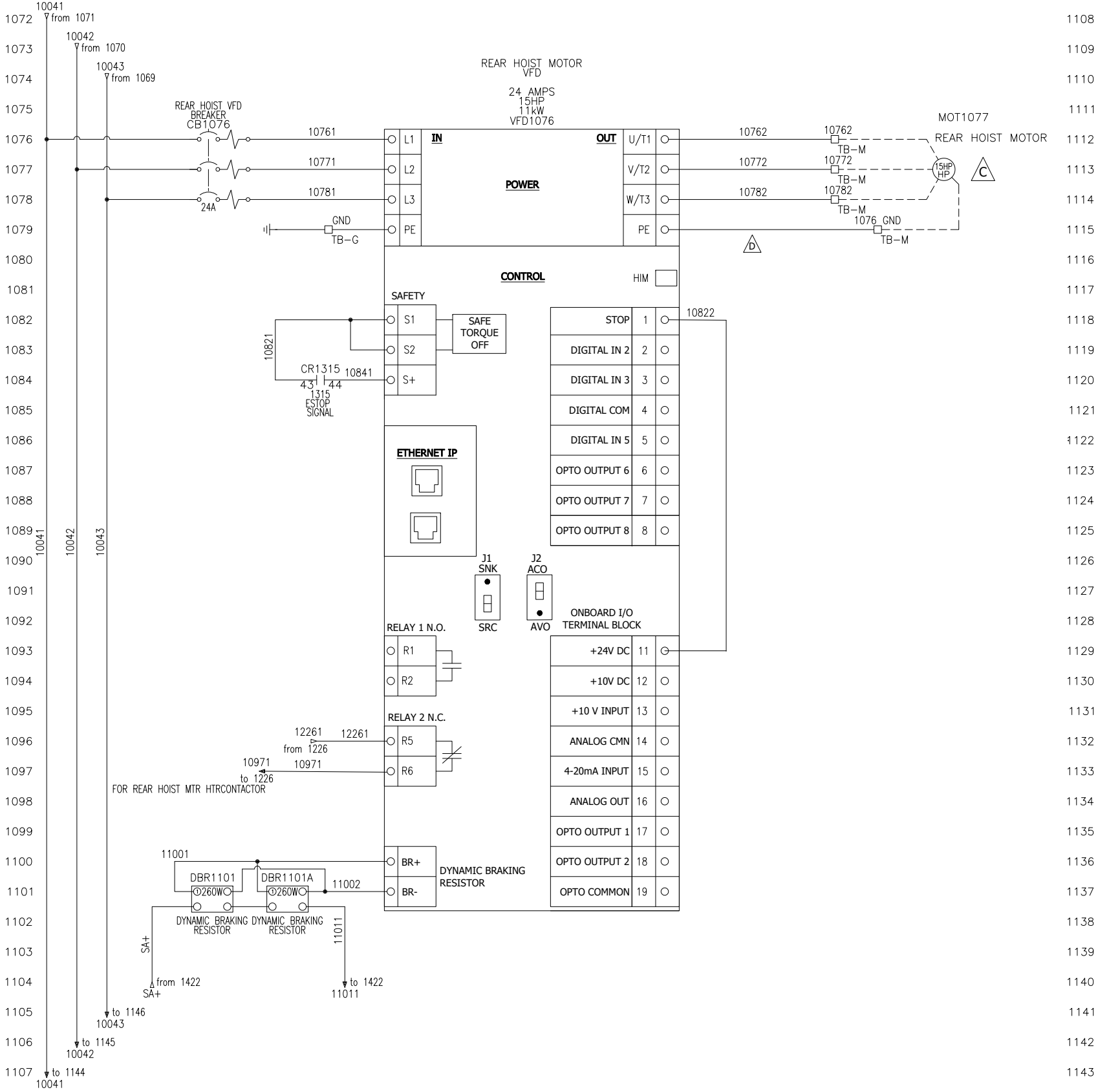


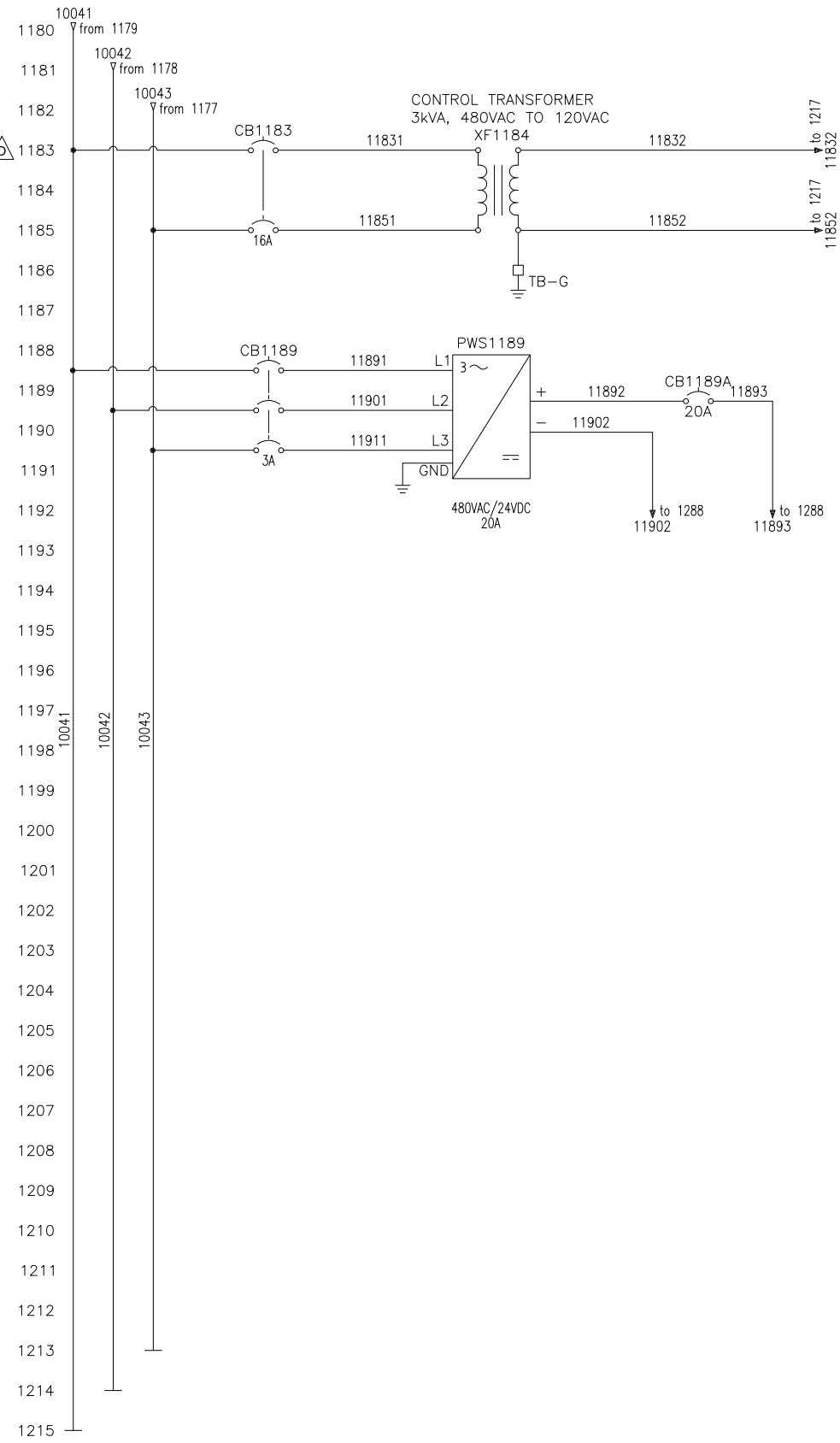
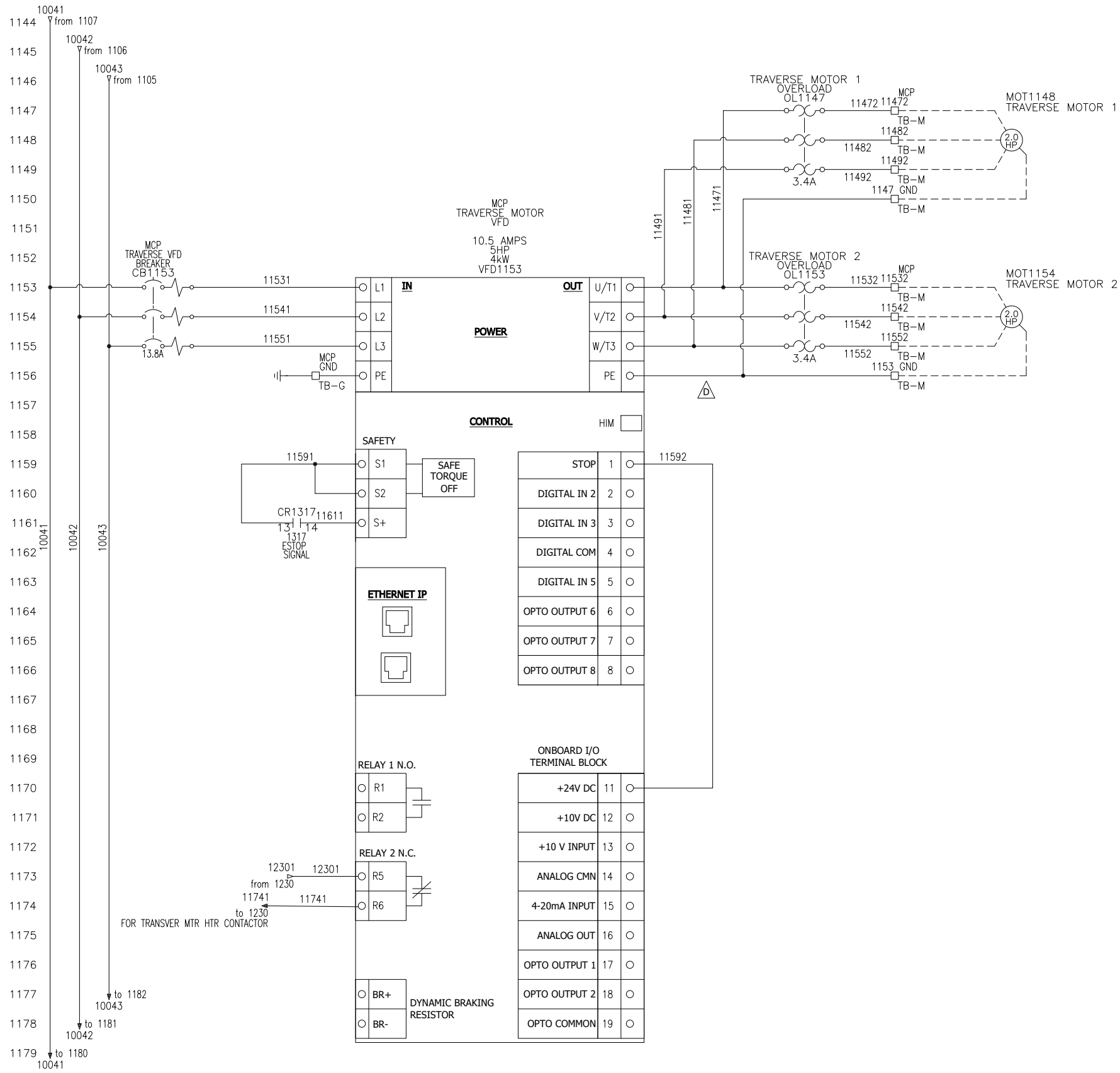
51	PLC1344	1	5069-RTB4-SPRING	ALLEN BRADLEY	REMOVABLE TERMINAL BLOCK FOR CONTROLLER MOD POWER, 4-PIN
52	PLC1344	1	5069-RTB6-SPRING	ALLEN BRADLEY	REMOVABLE TERMINAL BLOCK FOR CONTROLLER SA POWER, 6-PIN
53	PLC1362, PLC1398, PLC1434	3	5069-IB16	ALLEN BRADLEY	COMPACTLOGIX 16-POINT DIGITAL INPUT MODULE, 24VDC INPUTS
54	PLC1470	1	5069-OB16	ALLEN BRADLEY	COMPACTLOGIX 16-POINT RELAY OUTPUT MODULE, 24VDC OUTPUTS
55	PLC1545	1	5069-IY4	ALLEN BRADLEY	COMPACTLOGIX 4-POINT ANALOG INPUT MODULE, CURRENT/VOLTAGE
56	PLC1362, PLC1398, PLC1434, PLC1470, PLC1545	5	5069-RTB18-SCREW RTB	ALLEN BRADLEY	TERMINAL BLOCK, SCREW TYPE, 18 PINS, FOR ALL ABOVE PLC I/O MODULES
57	PLC ACCESSORY	1	5069-ECR	ALLEN BRADLEY	COMPACTLOGIX RIGHT END CAP FOR TERMINATION
58	HMI1331	1	2711P-T15C21D85	ROCKWELL AUTOMATION	PANELVIEW PLUS 7 STANDARD TERMINAL, TOUCH SCREEN, 15 INCHES, TFT COLOR, SINGLE ETHERNET, 24V DC, WINDOWS CE 6 , STANDARD MODEL
59	ETH1333	1	1783-CMS20DN	ALLEN BRADLEY	STRATIX 5200, 18 COPPER 10/1000 PORTS, 2 COMBO 10/1000 PORTS
60	WC1335	1 SET	FLEX 8EX2 SERIES	MAGNETEX MATERIAL HANDLING	WIRELESS PENDANT SYSTEM, INCLUDING ONE RECEIVER AND TWO TRANSMITTERS, WIRELESS ANTENNA, CUSTOM LABEL, E-STOP, ONE SELECTOR
61	CR1245	1	RH3B-ULC-120AC	IDEC	CONTROL RELAY, 3PDT, WITH INDICATOR AND CHECK BUTTON, 120VAC
62	CR1245	1	SH3B-05C	IDEC	SOCKET, RELAY
63	CR1470, CR1472, CR1480, CR1482, CR1484	5	RH3B-ULC-DC24V	IDEC	CONTROL RELAY, 3PDT, WITH INDICATOR AND CHECK BUTTON, 24VDC
64	CR1470, CR1472, CR1480, CR1482, CR1484	5	SH3B-05C	IDEC	SOCKET, RELAY
65	PB1364	1	ABD111NUB	IDEC	PUSH BUTTON - MOMENTARY, NEMA 4/4X, BLACK, WITH 1 N.O. + 1 N.C. CONTACT BLOCKS
66	LT1259	1	APD1QH2DNW	IDEC	WHITE PILOT LIGHT - STANDARD, FULL VOLTAGE, NEMA 4/4X
67	PB1362	1	AVD311NUR	IDEC	MUSHROOM HEAD EMERGENCY STOP BUTTON, 40 MM, PUSH LOCK, TURN RESET, RED, NEMA 4X, WITH (1) N.O. + (1) N.C. CONTACTS
68	SS1390	1	ASD211NU	IDEC	TWND SERIES, 2-POSITION MAINTAINED SELECTOR SWITCH, KNOB TYPE, WITH (1) N.O. + (1) N.C. CONTACT BLOCKS
69	PB1364, LT1259, PB1362, SS1390	4	GRAVOGRAPH	GRAVOPLY ULTRA	GENERIC OPERATOR TAG - ENGRAVE AS SHOWN
70	TB-M	13	3038338	PHOENIX CONTACT	3LEVEL TERMINAL BLOCK - ST 4-PE/3L, MULTI-LEVEL, 32AMPS, GRAY, 0.08-6MM^2, 28-10 AWG, CLIPLINE - SPRING CAGE TERMINAL BLOCK
71	TB-GND	10	3044759	PHOENIX CONTACT	GROUND TERMINAL, NUMBER OF CONNECTIONS: 4, CONNECTION METHOD: SCREW CONNECTION, CROSS SECTION: 0.14 MM2 - 6 MM2, MOUNTING TYPE: NS 35/7.5, NS 35/15, COLOR: GREEN-YELLOW
72	TB-1, TB-2, TB-3, TB-D1, TB, TB-AI	70	3044814	PHOENIX CONTACT	DOUBLE-LEVEL TERMINAL BLOCK, NOM. VOLTAGE: 800 V, NOMINAL CURRENT: 30 A, CONNECTION METHOD: SCREW CONNECTION, 1ST AND 2ND LEVEL, RATED CROSS SECTION: 4 MM2, CROSS SECTION: 0.14 MM2 - 6 MM2, MOUNTING TYPE: NS 35/7.5, NS 35/15, COLOR: GRAY
73	TB-1, TB-2, TB-3, TB-D1, TB, TB-AI	10	3047293	PHOENIX CONTACT	END COVER
74	TB ACCESSORY	30	800886	PHOENIX CONTACT	END BRACKET - E/NS 35 N, ACCESSORY, END BRACKET, GRAY, FOR THE NS 35 DIN RAIL, 9.5MM WIDTH
75	ACCESSORY	AS REQUIRED	807012	PHOENIX CONTACT	DIN rail perforated, acc. to EN 60715, material: Steel, galvanized, passivated with a thick layer, Standard profile, color: silver, 35mm, 7.5mm, 1000mm, DIN
76	ENCLOSURE ACCESSORY	A/R	TYD2X3NPW6	THOMAS & BETTS	NARROW SLOT WIRING DUCT, PVC, 2"x3" HIGH
77	ENCLOSURE ACCESSORY	A/R	TYD2CPW6	THOMAS & BETTS	WIRE DUCT COVER FOR 2" DUCTS
78	ENCLOSURE ACCESSORY	A/R	TYD1X3NPW6	THOMAS & BETTS	NARROW SLOT WIRING DUCT, PVC, 1"x3" HIGH
79	ENCLOSURE ACCESSORY	A/R	TYD1CPW6	THOMAS & BETTS	WIRE DUCT COVER FOR 1" DUCTS
80	ACCESSORRY	1	679002	OVIVO	NAME PLATE (ETCH AS SHOWN)
81	ACCESSORRY	1	49046A	OVIVO	NAME PLATE - "DANGER"
82	"TRASH RAKE MAIN CONTROL PANEL"	1	592273	OVIVO	ENCLOSURE EQUIPMENT TAG ID, BLACK BACKGROUND WITH 3/16 INCH WHITE LETTERS, NOMINAL HEIGHT OF NAMEPLATES 2 INCHES.
83	GROUND BAR	1	GBK520	EATON	5 TERMINAL GROUND BAR KIT,1-3/4 IN MOUNTING HOLE DISTANCE, 1-2/0 LUG,GROUND BAR KIT,CH,5 TERMINALS,0.75 IN,CH LOADCENTERS, FOR 2/4 CIRCUIT LOADCENTERS USE TYPE GBK5 OR GBK520 GROUND BAR
84	CRSAFETY1310	1	LG5925-48-61-24	AUTOMATIONDIRECT	DOLD SAFETY RELAY, EMERGENCY STOP AND SAFETY GATES, 2-CHANNEL, 24 VAC/VDC, (3) N.O. SAFETY OUTPUT(S), (1) N.C. MONITORING OUTPUT(S), FIXED SCREW TERMINALS.
85	CR1315, CR1317	2	AFS09-30-22-11	ABB	AFS SERIES, ABB SAFETY CONTACTOR SERIES AFS09-30-22-11, 20-60VDC
86	TB-M	4	3044102	PHOENIX CONTACT	UT-4 FEED THROUGH TERMINAL BLOCK NOM. VOLTAGE 1000V, CURRENT 32A, NO. OF CONNECTION 2, SCREW CONNECTION, CROSS SECTION 0.14MM-6MM, MOUNTING TYPE NS 35/7.5, NS 35/15, GRAY COLOR
87	TB-M	4	3047028	PHOENIX CONTACT	D-UT 2.5/10 END COVER

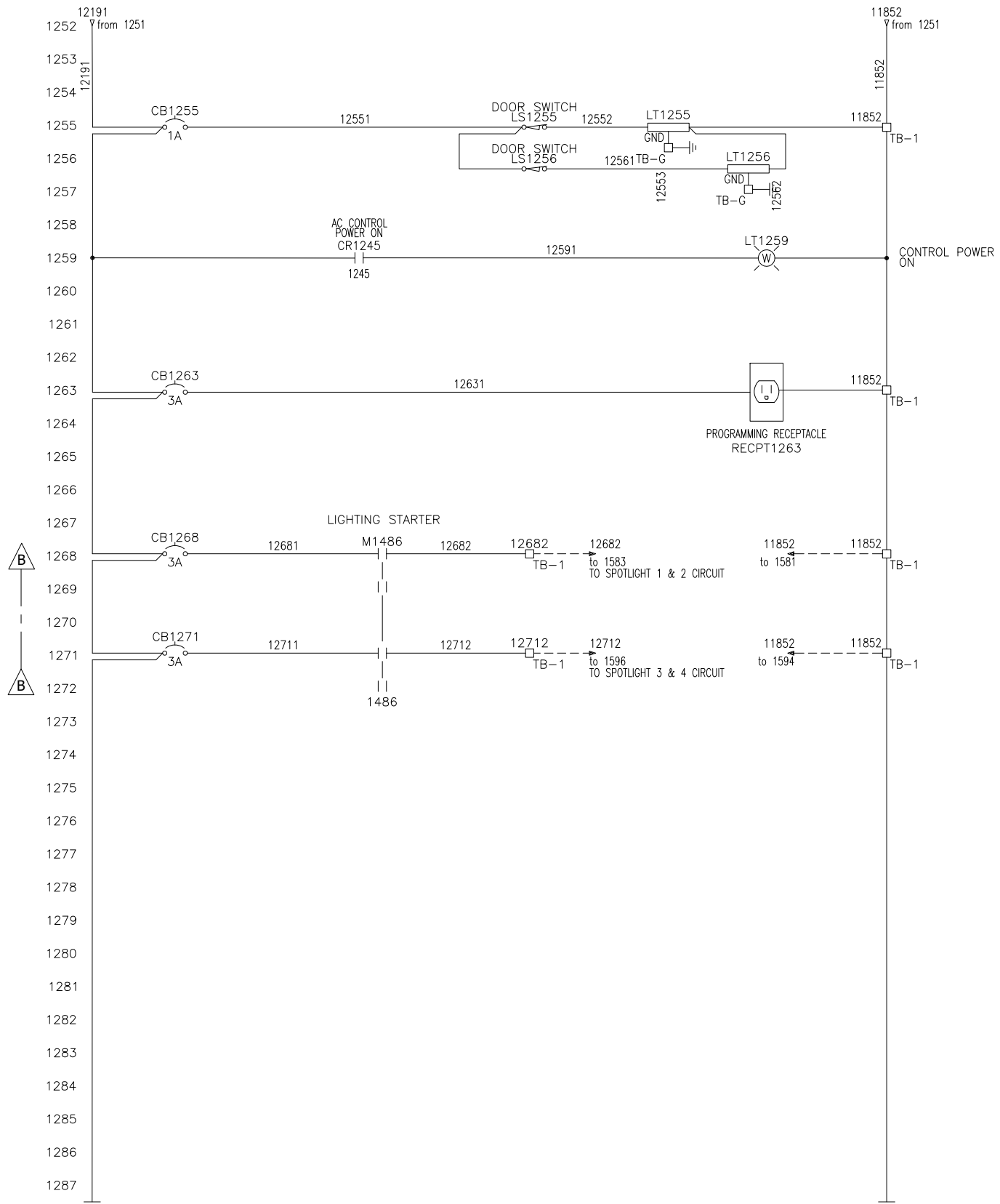
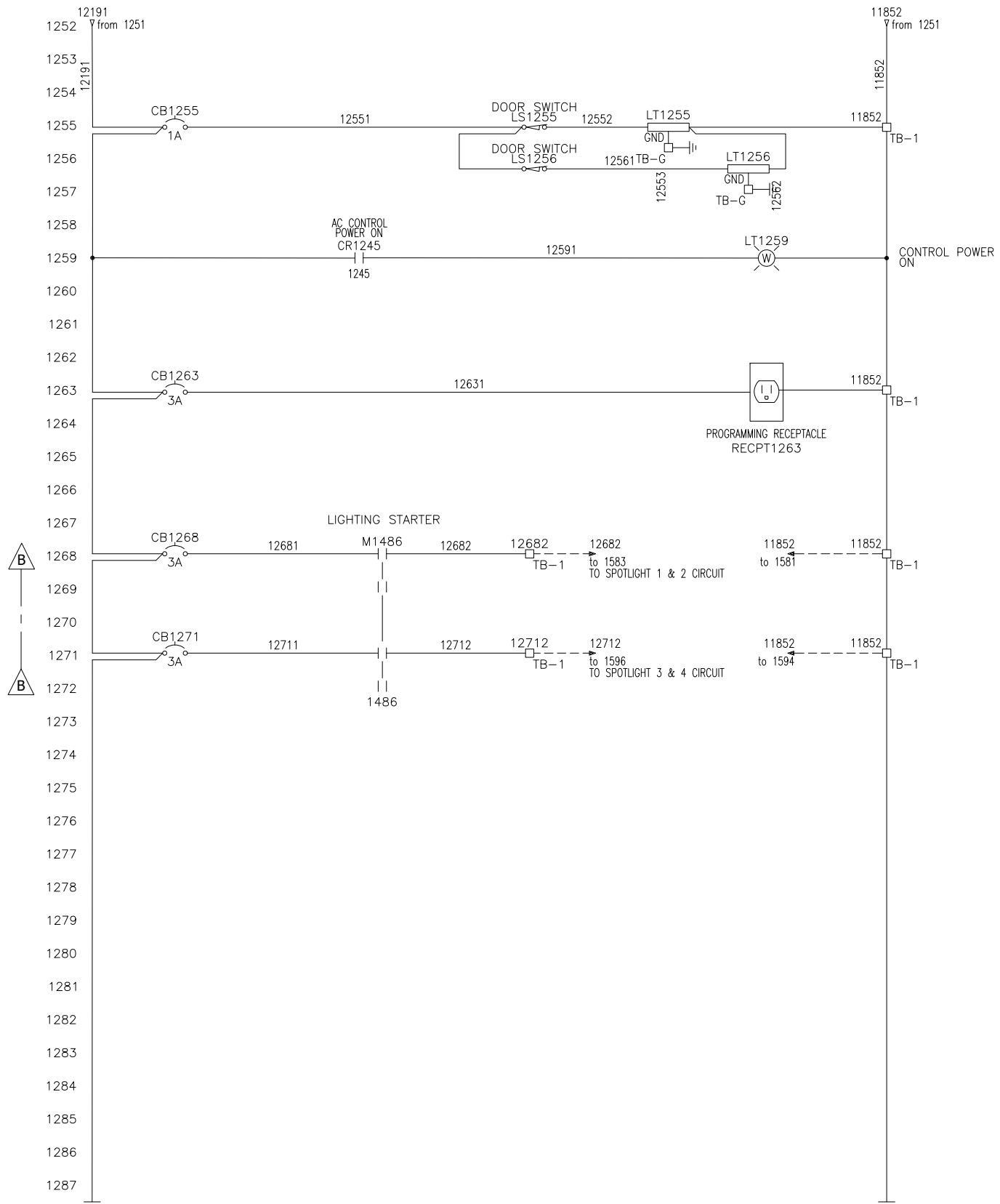


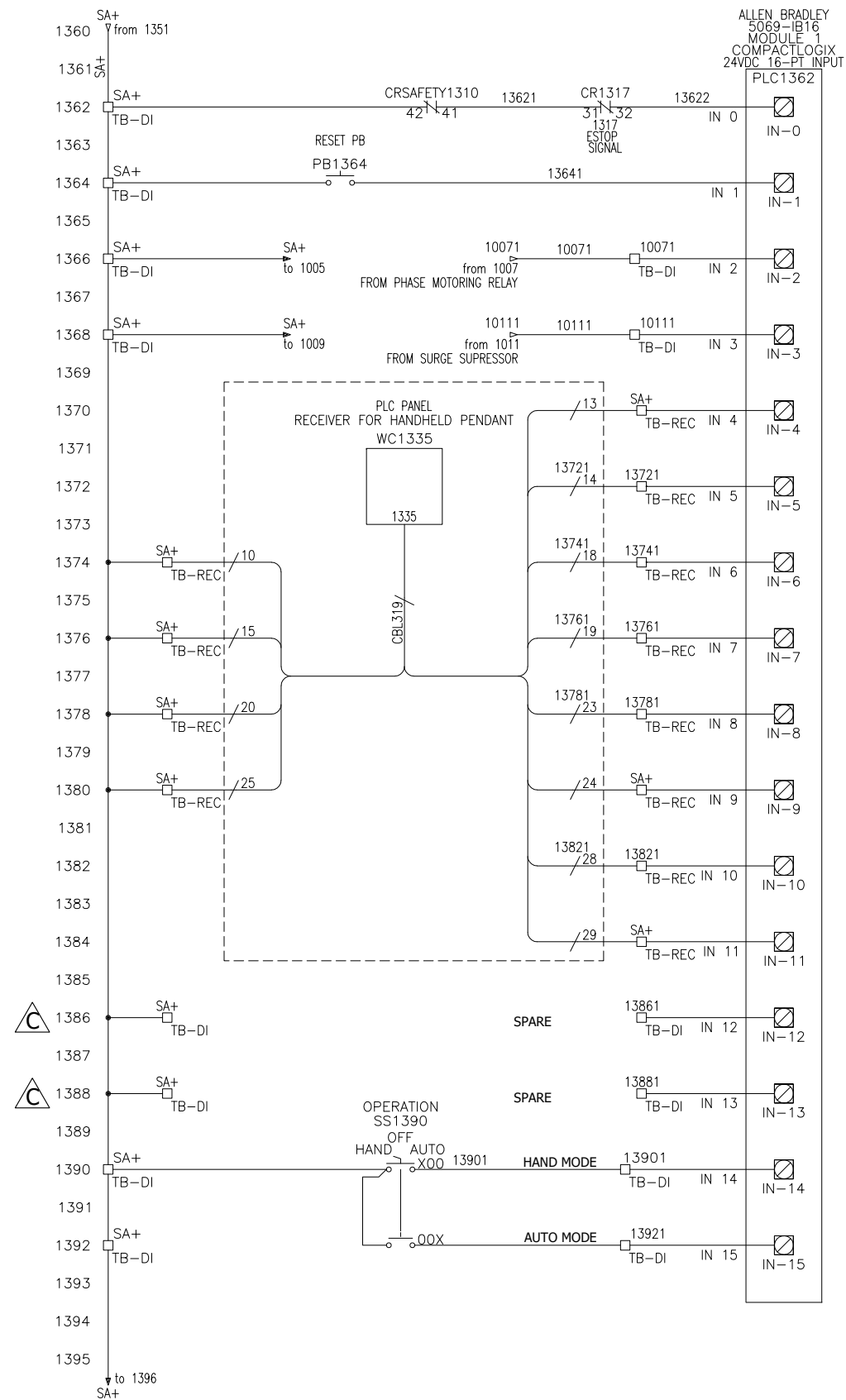
480 VAC, 3-PHASE, 60HZ
POWER SUPPLY INPUT FROM CUSTOMER











HOIST UP DIRECTION

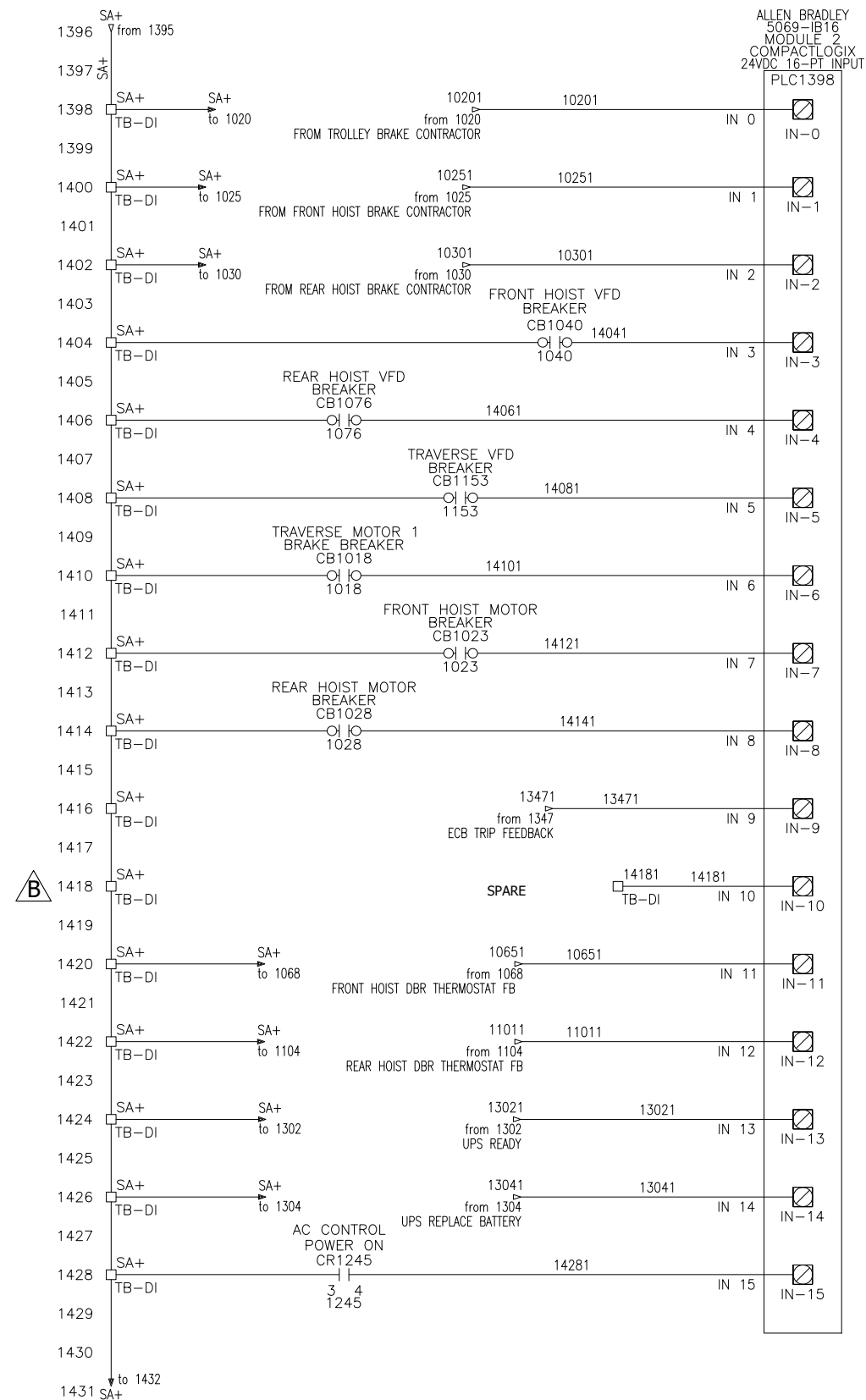
HOIST DOWN DIRECTION

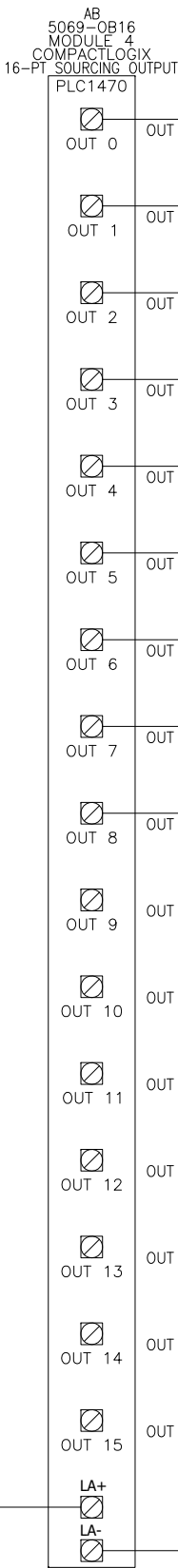
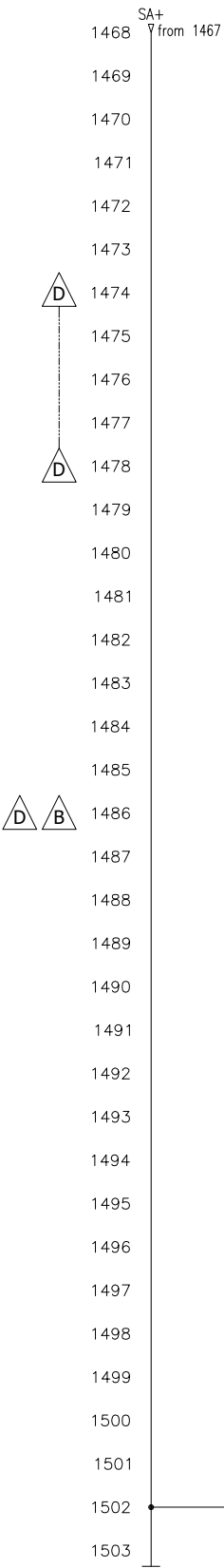
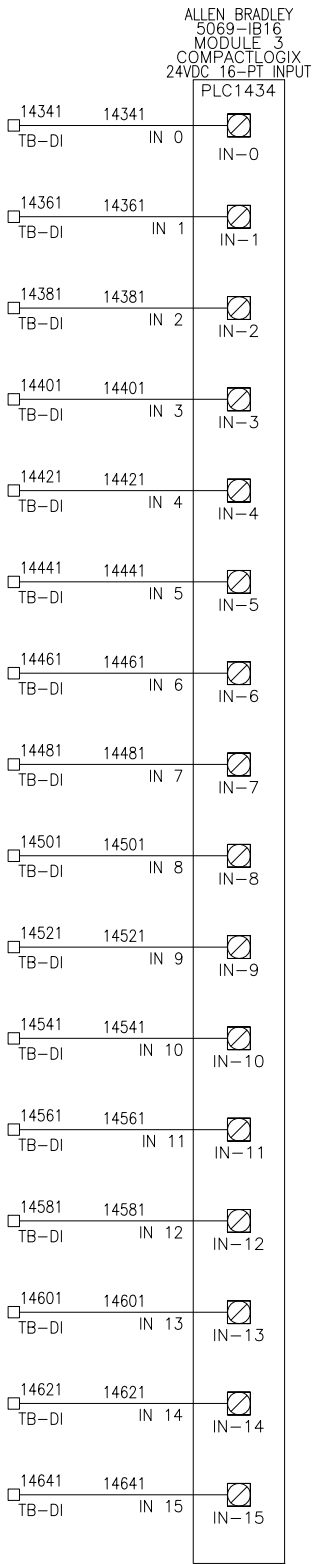
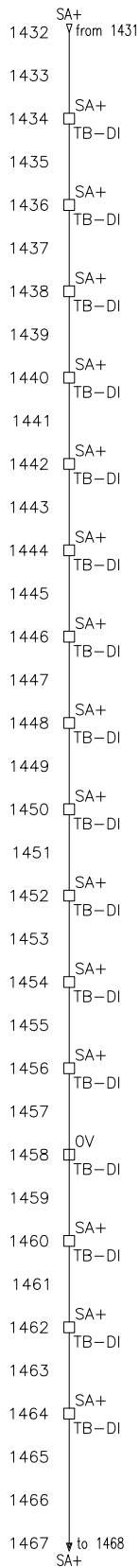
TROLLEY FORWARD DIRECTION

TROLLEY REVERSE DIRECTION

RAKE HOIST OPEN

RAKE HOIST CLOSE



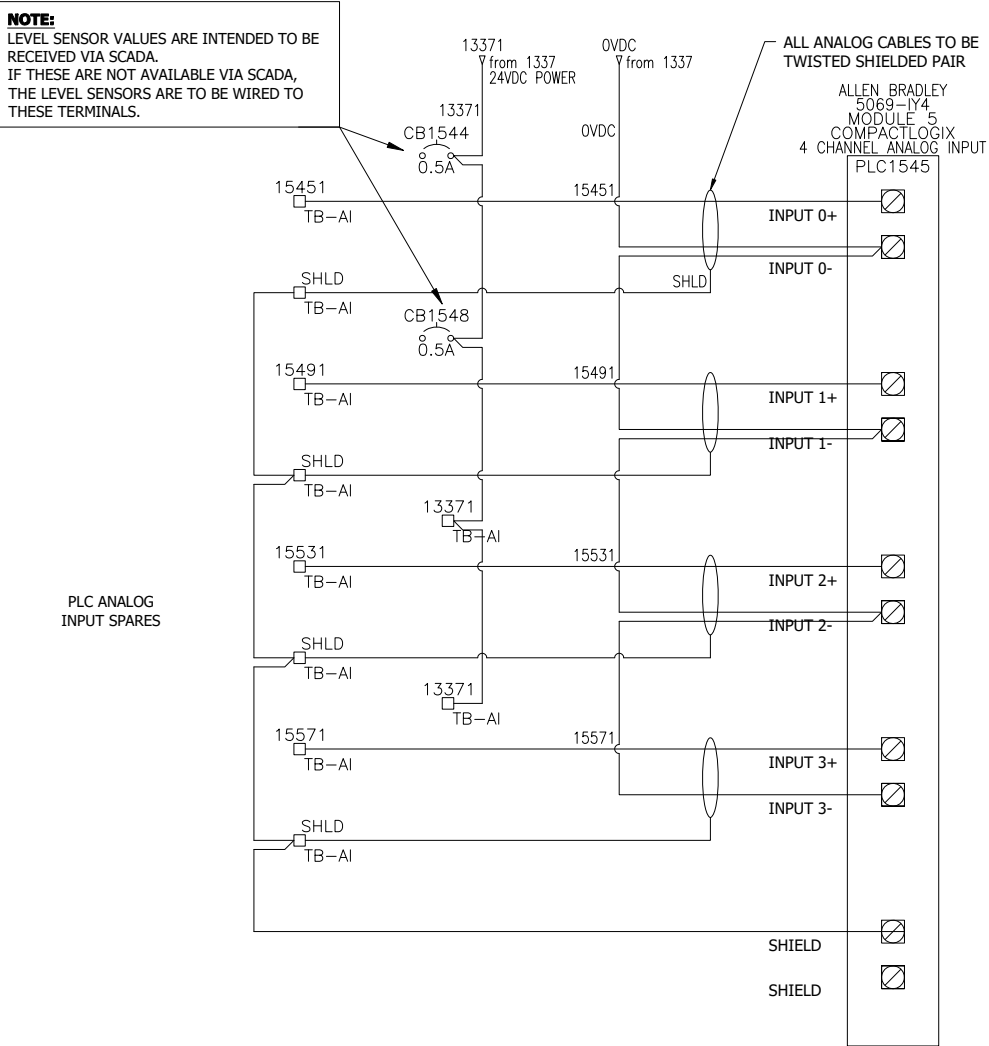


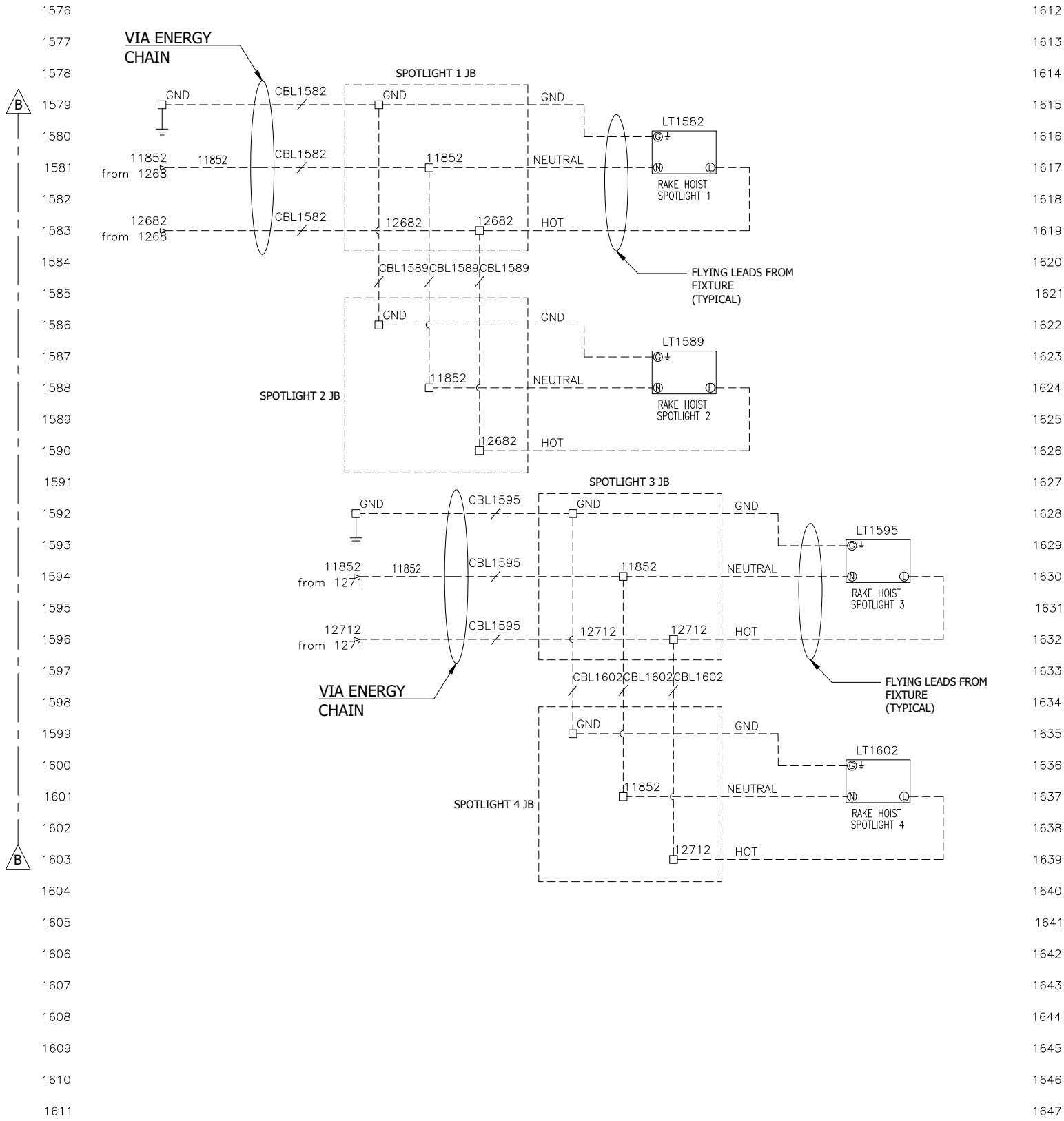
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